

Prague University of Economics and Business
Faculty of Informatics and Statistics



**Statistical analysis of situations in
sport**

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Author: Viliam Virba

Supervisor: Ing. Jakub Danko, Ph.D.

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Abstract

Bachelor thesis offers insight into the specific sport situations from the statistic perspective that is possible due to availability of large amount well structured data that measure many aspects of the games. Every situation is present in American football and data are exclusively sourced from National Football League since 2000. American football as a sport is introduced followed by dataset description. After the league itself and dataset is described three hypotheses are formulated. First is trying to show how aggressive tactics could consequently lead to losing in tight games. Second hypothesis looks at the frequent event that is taking a timeout before opposing team is trying to score an important points by kicking a field goal. Last third hypothesis is trying to measure effect of stadium altitude on distance that is possible to be kicked by punter. Hypotheses are evaluated by statistical hypotheses testing after data are processed in desirable conditions.

Keywords

American football, NFL, sport statistics, sport analytics, hypothesis testing, data driven decision making

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Introduction

Since the very first days of all sports there has always been at least notation of occurrences counts in order to determine frequency of some specific situations. Even number of total goals or points is an example of descriptive statistics that can lead into frequentist probability dimension. When technology allows us to measure, collect and store large amounts of data it was inevitable that data analytics will revolutionise the world of sports. Through well structured data analyses from every game it is possible to make adjustments in every possible aspect of the sports. Outputs are specifically prepared by data analytic departments in every professional team and information extracted from the data then can help coaches and even individually every player to make decisions with objective of higher probability of success.

One of the most strategic sport with multiple aspects where decision making in a real time takes place is American Football. It is not a coincidence American Football is considered to be a chess with the athletes. Overall strategy and tactical manoeuvres importance with possibility to influence every game in real time from the outside by adjusting plays brought American football into becoming one of the sports where data analytics is most important and applied in these days. This thesis focus on the NFL data and analysing specific situations in games that are standardised and replicable. Standard situations include mainly punts, field goal attempts or even effort to convert a fourth down. Good specification in combination with available data that can be processed allow exploration of game aspects with possible conclusions that can have advisory character or can at least be giving feedback to the teams. Three such hypothesis are formulated and included in this thesis in order to show how search for the answers can be achieved with good questions being asked. First is description of venturesome decision that can result into losing a tight game. Second hypothesis is focused on regularly used technique to lower chances of opposing team to score. Last third is exploring how external factor if altitude can influence flight properties of ball and results of the kicks.

1. American football

1.1 Introduction to National Football League

„The National Football League (NFL) is a professional American football league that consists of 32 teams, divided equally between the American Football Conference (AFC) and the National Football Conference (NFC). The NFL is one of the major professional sports leagues in the United States and Canada and the highest professional level of American football in the world.“ (1)

Currently National Football league has biggest market share from all North American most famous professional leagues in United State such as National Hockey League, Major League Baseball or National Basketball Association league. Final round of playoffs called the Super Bowl is big cultural event in United States every year and is one of the globally most watched events in sports. Winner of this game is crowned as World Champion annually with the Vince Lombardi trophy.

1.2 Football vocabulary

Possession means that the team has a ball in their hands and thus can attack trying to score a touchdown by gaining more and more yards in field position. Every game is divided into drives which are basically different possessions of both teams. End zone is name of the area that is at the both ends of the 120 yards long field. Drive is starting either by kickoff on start of the both halves (overtimes) or after a scoring some points by either of the teams and eventually by punt which is present in the case when punting team does not score any points but do not want to take a risk of losing a possession and giving the opponent a good position on the field. Field goal is one of the objectives for a team that is awarded with less points than main objective that is touchdown awarded with double of 3 so 6 points with an opportunity to kick from a very short distance for 1 point called point after touchdown. Field goal territory is the area from which it is possible to successfully attempts a field goal. Fourth down territory is the area where team with a possession is not close enough to be in a field goal territory and at the same time too close to the end zone so the punt is not providing any advantage. Converting a fourth down means that if team take a risk of playing on last attempt and gets over 1st down line by that one play they can continue in a current possession so the drive continues. When a team do not manage to get behind line of the first down on 4th attempt, automatically losses a possession because of turnover on downs with fourth down not converted. Two minute warning is so called referee timeout that stops the game clock with two minutes left in each of the halves. Safety is a possible scoring situation of opposing team when play ended in the own end zone of the team with possession and is awarded by two points to the opponents associated with change of possession.

1.3 Typical match

In general game of American football is divided into two separate halves and four quarters with duration of fifteen minutes. Match consists of offensive drives for both teams which are built by multiple plays. Every half is started by kickoff to determine starting position of the offensive drive while quarters are just for changing sides while the drive continue.

Offensive plays can be categorised as mostly run or pass with an objective of gaining as much yards as are possible vertically in direction to the opponent end zone. Every drive is alive while offensive team is able to progress on the field by at least 10 yards from maximal 4 attempts which are called downs. Typically team is attacking by offensive play till is able to manage 10 yard distance progress from 3 downs.

When a offensive players did not manage to move the ball from the 1st down line of scrimmage to position that is longer than 10 yards in 3 attempts 4th down dilemma comes into the account. Usually when team is not able to kick a field goal thus score three points because is not close enough to the end zone of defending team coaches decide to punt a ball. It is with an intention to worsen field position of the opposing team at the start of their offensive drive as much as possible. This aspect of play is often referred as battle of the field position. After kicker punts the ball opposing team has opportunity to return the ball to enhance their field position. When return is ended the line of scrimmage is determined and the receiving team of the punt now has their first 4 downs to gain 10 yards of field position and get new set of 4th downs when they convert one of the downs. Main goal is to score a touchdown what means that offensive player either penetrate the end zone of opposing team with a ball in possession or catches a ball into the end zone.

Teams alternate in the possession and drives as they either score by touchdown or field goal or loss a possession due to giveaway or punt. Giveaway can be forced by fumble, interception and turnover on downs. Fumble is a situation when player loss a possession of ball usually due to contact with opponent before he is down by contact on the ground. Interception is when usually forward pass of quarterback is being caught not by the receiver but by any member of the defense. Turnover on downs occurs when a team did not manage to convert 4th down and gives a ball to opponent on the yard line 4th down play ended. At the end of the regulation which is at the end of the 4th quarter wins the team that has scored more points.

If the scoreboard is tied game continues into the overtime. Rules of overtime are slight different in NFL and other leagues. Currently in the National Football League regular season if the overtime of 10 minutes did not determine a winner the game is considered as a draw. Playoffs need to have one winning team that can advance into another round so the overtime is played till the match is decided. Current overtime rules are that first possession determines the coin toss. First touchdown is a decider and team that scores first wins the game. When in the first possession field goal is scored second team has chance to at least tie with field goal or naturally decide with a touchdown. After this two first possession of each team every other scoring decides the match and determine winner.

1.4 Individual positions

In all of three sub teams there are very special types of the players on different positions. Offensive plays are usually either run plays or pass plays with one forward pass allowed. Quarterback is a player that is director of the offence and the one who is usually throwing a ball forward. Wide receivers are players that are trying to catch forward passes by Quarterback. Runningbacks are players that gets ball usually from the Quarterback in the backyard and are trying to run straightforward mainly in the routes that blocking players creates for them, with intention to gain maximum count of yards, so they can move the ball furthest from the line of scrimmage. Tight ends are the players that can block and by blocking protects quarterback or create a route for running back but also can catch a ball behind the line of scrimmage in forward direction. These positions vary by different schemes and formations of the offense. Players that usually not vary much are the offensive line men. Offensive line consists of the centre that snaps the ball at the start of every offensive play and two so called guards on the interior of the line and two offensive tackles on the exterior. Main role of the offensive line is either quarterback protection in mostly passing plays and creating routes for running backs during run plays.

Defense is not so segmented as offence but also there are multiple different positions and has 3 layers. Defensive line is the first layer that is closest to line of scrimmage and consists of usually 4 players. Defensive tackles are left and right on the interior of defensive line and their task is to block running attacks. Defensive Ends are also left and right on the exterior of defensive line and their task is to get onto passing player preventing the pass to be executed and can also stop some exterior runs. Second layer are linebackers that are three, two on each side left and right outside linebackers and the middle linebacker. Linebackers task is primarily to stop the runs but can also cover potential receivers of the pass so they are something in between first and last layer. Last third layer consists of two cornerbacks and two safeties. Cornerbacks are on the exterior of the field covering receivers that are near both sides of the field and try to cover them so catching a ball is not possible. Safeties are the last instance that can prevent deep pass and they are usually behind and between cornerbacks.

Special teams positions are mainly related to act of kicking a ball. Punter is kicking a ball how far it is possible after unsuccessful drive. Kicker is kicking field goals, extra points after touchdowns and kickoffs while kickoffs can handle also kickoff specialist that is only specialised in kickoffs. Then there is a squad that helps with realisation of field goal attempts, points after touchdown and punts. Squad consists of longsnapper what is basically a center snapping a ball directly to punter in case of punt or into the hands of holder who holds the ball for the executor of the field goal or point after touchdown kick. Kicker or punter returner is a player specialised in returning kicks who try to similarly as running back by his agility to bring ball as further as possible. Gunners are trying to stop a returner and are blocked by the Jammers that are creating a route for the returner. Upbacks are something as safeties in return situations that prevent returner to score a touchdown from the return as the last instance.

1.5 Specific situations

Fourth Down is first situation that is being analysed. Every team offensive players has 4 attempts of plays to acquire at least 10 yards cumulatively. Plays that count are marked as downs and are a way how to move the line of scrimmage that defines start of the following play. Conversion means that 10 yards were gained by maximum of four downs. Objective is to score a touchdown what means to penetrate end zone with a ball in possession or catching a ball into the end zone. On the way to the end zone offensive players can face terminal situations for their drive. From the strict defined situation that terminates the drive and it is not converted 4th down or takeaway of the ball by opposing team to the very important 3rd downs. Prevention for not converting a 4th down thus not scoring any points with necessity to give a ball into the opponents hands is not to play 4th down after the 3rd down was not converted. Trying to convert a 4th down is kind of gamble that should be done only in specific circumstances because losing of potential 3 points can be crucial at the end of the match.(2)

Field goal is second specified situation. Every offense has more than one option of scoring a points in game of American football. Touchdown is a primary objective since is awarded with 6 points with another option of 1 point point after touchdown alternative or a 2 point conversion. Two point conversion is basically 4th down that need to be converted at the 2 yard line before end zone in order to get 2 additional point to touchdown 6. In comparison with point after touchdown it is more risky as it is in general harder task to execute. Field goal is an alternative to touchdown on how to score a points in a drive where usually team need to convert 4th down of any length to be able to continue and not punt the ball away. When a team is in field goal territory what means that field goal attempt is possible to be made it is considered conservatively rational to kick a field goal and score 3 certain points from that drive and not to risk loss of the ball with not converted 4th down with zero points acquired from that drive. Point after touchdown is basically a field goal from short distance that has very big probability of being successful. Scoring from the field goal attempt can often mean the difference between win and loss. Teams are understandably trying to use any possible legal method to prevent opposing team to make a field goal good. One of the method that is also last possible chance is freezing the kicker about which is the hypothesis related to field goal attempts.

Punts are last standardised situation described in the last hypothesis. Punting a ball is an act when usually a kicker of the team which evaluate a try of the 4th down conversion as too risky is trying to push opponents team as far as possible from current line of scrimmage. In other words from place where next play of opposing team would be played in case of turnover on downs caused by not converted 4th down risk. It differs from kickoff because in case of punt kicker have to kick the ball that is released from his hands before the ball touches the ground in the contrast when during a kickoff the ball is whole time placed on the ground. Because of this factor main target for the punter is to kick a ball as far as possible to worsen field position on the start of the drive of opposing team as much as possible. In specific cases when kicker is in shorter distance where it is not problem for him to kick a ball behind the

end zone also accuracy plays a role. When ball is kicked behind the end zone it means a touchback and start of the drive from 20 yard line thus kicker is trying to place a ball at least between 20 yard line and end zone as close to end zone as it is possible. Because of this aspect of kicking a ball into the longest distance possible in cases when it is almost impossible to kick a ball behind the end zone there is an option to explore flight properties of the ball what is used in following hypothesis of how the altitude influence flight properties of the ball.

2. Data and statistic methods

2.1 Dataset description

Dataset provide access to all games played in National Football League in 3rd Millennium so far thus from season 2000/2001 to last completed season in the time of writing bachelor thesis 2022/2023. Professional football season in United States starts at September and ends in February. Data is structured chronologically as play by play Panel Data through all these games from last 22 seasons with more than 1 100 000 observations each in separate row and 385 columns with granularity of every event occurred during a match. Columns are representation of game aspects, some original measured or written down and some as derived implicitly from original ones. Through the analysis there is a possibility to derive more new columns for better work with the dataset.

Across the columns it is possible to spot almost all kinds of datatypes. There are most basic binary boolean indicators as `timeout` or `touchback` where 1 or True states occurrence of timeout or touchback during that play and 0 or False states the absence. Categorical variables as `field_goal_result` with in this case 3 possible outcomes of made, missed, blocked or `penalty_type` with numerous of categories. Discrete numeric variables as `kick_distance` by 1 yard bounded from 0 to 100 yards and unbounded as `total_yards` where 1 yard is the smallest difference but theoretically can go both directions up to infinity and negative infinity. `Home_score` is example of integer where smallest difference is 1 point and values can be only positive natural numbers. Continuous numeric variables as WPA Win probability added within interval of $<0-1>$ and unbounded EPA (Expected points added) `total_home_pass_epa` which are examples of float datatype. There are also text or string variables as `td_player_name` which are unique thus can not be categorised. `Time_of_day` in format of hours, minutes and seconds is date time datatype example.

Most important columns that are used in analyses later are `game_id` and `play_id` unique to every game and every play respectively. `Play_type` column states what kind of play happened from the perspective of team with a possession. `Timeout` column determines whether there was called a timeout on that play or was not. `Posteam` indicated as string variable that can be converted into categorical variable since we have stable small number of unique teams (currently 32). `Qtr` is a category in what quarter happened the play. `Field_goal_result` is categorical variable with 3 possible outcomes one indicating scoring points labeled “made” and two other of not scoring points when field goal was either missed or subcategory of missed is blocked when opposing team managed to block a kick. `Score_differential` is difference in points scored till the time of the play. `Half_seconds_remaining` are seconds left in each of the halves. `Touchback` is binary indicator if the touchback occurred on that play. `Yard_line_100` gives information about line of scrimmage field position at the start of the play respectively to distance of the end zone for possession team. When a kick is practised in case of punts or field goals `kick_distance` states length of the kick as difference between line of scrimmage from start of the play and new line of scrimmage of following play.

2.2 Statistic methods

2.2.1 Student's t-test

„It is one of the most popular statistical techniques used to test whether mean difference between two groups is statistically significant. Null hypothesis stated that both means are statistically equal, whereas alternative hypothesis stated that both means are not statistically equal i.e., they are statistically different to each other“.(3) Every t-test is following Student's t-distribution and is a parametric statistical hypothesis test. T-test can be in a version of one sample test when location of parameter most frequently the mean in the t-distribution is being evaluated against null hypothesis. Alternatively there is an option of t-test when two independent groups are being compared against null hypothesis of equality.

Test statistic formula is in the form of :

$$t = \frac{Z}{s} = \frac{\bar{X} - \mu}{\frac{\hat{\sigma}}{\sqrt{n}}} \quad (2.1)$$

where,

t is the test statistic of t-distribution,

s is the standard error of the mean and scaling parameter for t-distribution,

Z is a difference between group mean $\bar{X}$ and hypothesized population mean μ ,

$\hat{\sigma}$ is estimate of population standard deviation,

n is the number of observations.

2.2.2 Pearson's chi-squared test

„The most commonly used statistical method for comparing proportions is the chi-squared χ^2 test (often termed 'Pearson's chi-squared' after the turn-of-the-century statistical pioneer Karl Pearson)“.(4) Test is being used usually to compare groups that differ in some category. Null hypotheses of the test is that proportion between groups are equal. Alternative hypothesis is that proportions are statistically not equal on the chosen level of signification.

Test statistic formula :

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} \quad (2.2)$$

where,

χ^2 is Pearson's cumulative test statistic chi-squared distributed,

n is number of observations,

O_i is number of observations from i-th group,

E_i is theoretical or expected number of observations for proportions to be same.

2.2.3 Levene's test for equality of variances

„Testing for scale (variance) heterogeneity, prior to the main inference of location (mean) parameters, is a common diagnostic method in linear regression to evaluate the assumption of homoscedasticity“.(5)

Homogeneity of variances is an assumption that needs to be fulfilled in statistical tests when pooled estimate needs to be used. Null hypothesis states that variances are equal between the groups. Alternative is not equality of variances between groups.

Test statistic W has approximately F distribution and can be calculated by formula :

$$W = \frac{(N - k)}{(k - 1)} \frac{\sum_{i=1}^k N_i (\bar{Z}_{i\cdot} - \bar{Z}_{\cdot\cdot})^2}{\sum_{i=1}^k \sum_{j=1}^{N_i} (Z_{ij} - \bar{Z}_{i\cdot})^2} \quad (2.3)$$

where,

N is total number of observations,

k is number of groups,

N_i is number of observations in i-th group,

Z_{ij} is a difference in absolute value between Y_{ij} what is measured variable of j-th observation in the i-th, group and mean or median of the corresponding i-th group,

$\bar{Z}_i$ is the Z_{ij} mean for each i-th group,

$\bar{Z}_{\cdot\cdot}$ is mean of every Z_{ij} .

3. Hypotheses

3.1 Fourth down played despite of option for field goal attempt

3.1.1 Specification

Trying to convert 4th down by pass, run or some trick play is taking the risk of giving opposing team the ball by turnover on downs. Offensive team should take this kind of risk usually only when there is no other option and team needs the drive to continue at every price. If the team is not in that need it is considered better to conservatively punt the ball in order to not give opponent a good field position and rely on the defense to get the ball into the hands of offense again possibly in better field position. Even Higher risk that could be considered as a gamble is trying to convert a 4th down in the field goal territory. On the other hand naturally there might be situations when possessing team should not stop its drive because there is a chance that they will not gain a possession again in the remaining time of the match. If score difference is bigger than 3 points against team in offensive drive and there is a potentiality of not receiving another opportunity to score, the risk becomes necessity and should not be considered as a gamble.

Over the last years number of 4th downs played with a standard pass, run or trick play in order to convert is growing. We can see that trend on the line chart where the frequency of 4th down played in relation to total number of plays played in each year is calculated. This ratio displays how often a 4th down is tried to be converted over the years.

One of the explanations that is offering is introduction of so called Next Gen Stats in NFL that is available to all members of team from head coaches through coordinators up to players that can see them on the tablets when sitting on the sideline.

One important component of Next Gen Stats are probabilistic models that calculates probability of winning the match in every second during the match and also shows multiple scenarios of probable outcomes. This models contains multiple parameters that are taken into account and are black boxes that gives number but there is no chance to comprehend reasons enough by a person in a real time. This additional information could cause that coaches are not taking conservative way but are trying to gain more possible points from the situations.

There is nothing wrong in this approach on large scales of events but in NFL mainly there are only few events that are actually happening with no big change in circumstances. Because of that phenomena it is interesting to look at the simplified specific situations such as 4th down conversion efforts in a field goal territory.

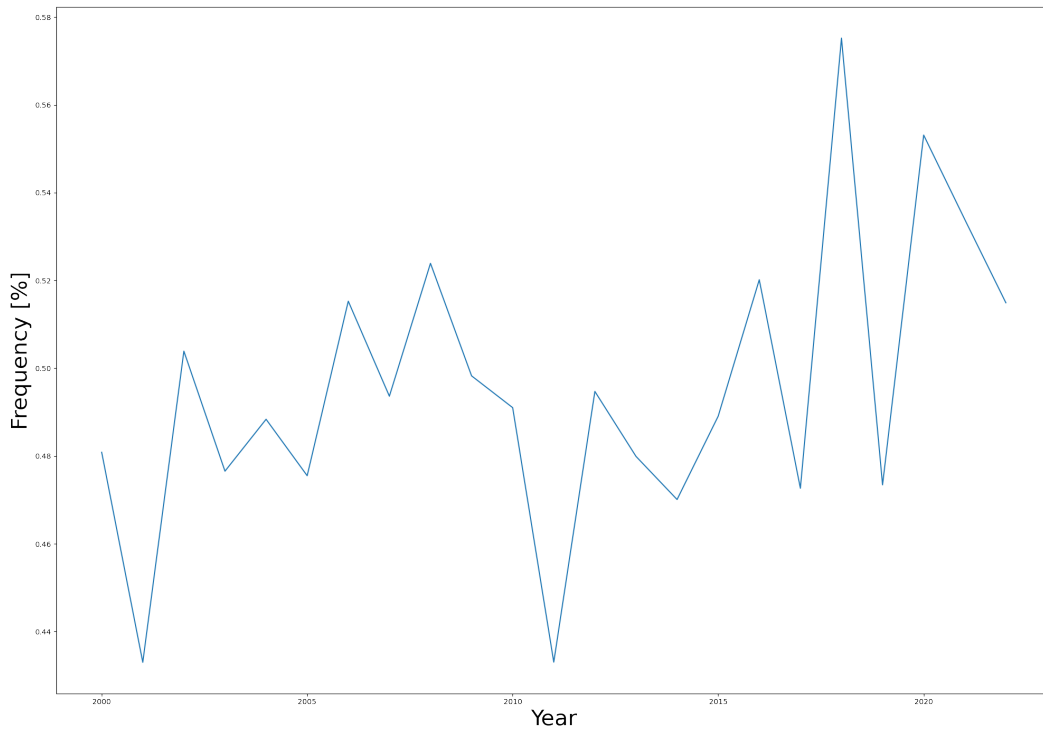


Figure 3.1: Frequency of played 4th downs over the years (own processing using Python)

3.1.2 Procedure

Converting a 4th down in the field goal territory is conservatively considered as an extreme gamble. Only in extreme situations such as mentioned before when it becomes a necessity it is rational to play it every time. It is a necessity when the team by stopping its offensive drive risks that offense will not get the ball again in the possession a needs to score more than 3 points to at least equal the scoreboard. In this specific case it is totally reasonable to go for the 4th down because team has practically no other option how to score necessary points when do not want to lose whole game.

What is more interesting is look at the situations where it seems like a gamble. It is because in that case team refuse to take 3 points and gamble with potential 7 points gain from 6 point touchdown followed by point after touchdown for 1 point. It is possible to look at the games where this phenomena occurred at least once. Filter it to games where taking 3 points in this situations could lead to at least equal the scoreboard at the moment of 4th down conversion attempt. For situations where more than once one team or another did not convert a fourth down in field goal territory we filter that game by number of 4th downs not converted in field goal territory multiplied by potential 3 points from the field goal attempt. From this potential score we can filter games where in reality losing team at the end of the regulation could have at least tie the game if this team would attempt a successful field goal. Since successful kick of the field goal is a condition to at least tie the game it is very important to set criterion of the probable field goal territory. In other words from what distance is very probable that field goal would be made. Best practise for not to set this criterion very strictly

is to look at the field goal distances distribution of made rates higher than 0.9. From data since year 2000 probability of making field goal attempt is higher than 0.9 from every field goal distance that is lower than 33 yards. This distance makes kick from the 23 yard line on the field and needs line of scrimmage field position at least on 23 minus 8 to be possibly executed. This territory will be considered as field goal territory criterion with more than 0.9 probability of scoring a points from the field goal attempt. By this strict restrictions there will be very filtered observations count but it will tell with a high certainty that this decision was a mistake. On bar chart can be seen displayed field goals distribution by every yard with brown proportion of made field goals, black of missed and purple of blocked. Light blue line that is integrated illustrate decreasing rate of made field goal attempts and yellow lines are locating place where is a boarder of 0.9 rate. Width of the bars represent frequency of attempts from each yard line.(6)

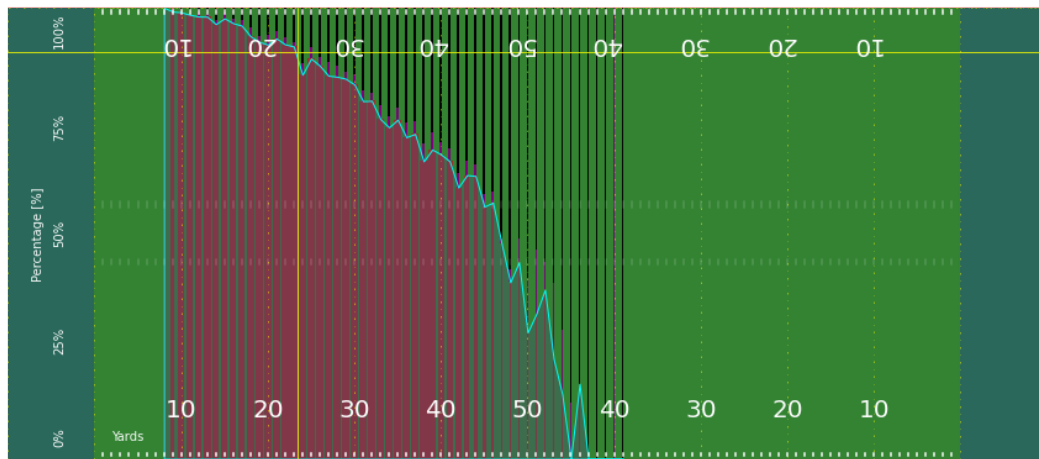


Figure 3.2: Probability of scoring from field goal attempt (own processing using Python)

Assumption of making a mistake in a decision process can be made because in this situation of kicking a field goal in both cases of making or not making team with possession would not give opposing team any big advantage of field position. In case of made field goal kickoff would let start opposing team on the ball in standard position somewhere around 30 yard line on own half of the field. While in case of not made field goal in this territory would give opposing team no large advantage. To sum up hypothesis is to look at how frequent not converting a 4th down in very probable field goal territory while at the moment of play score differential is lower than 3 points could influence the match result. As mentioned before we can make these direct assumption due to strict restrictions and almost independence on following events in the match caused by separation of each offensive drive. As a result important is a ratio of games that could have at least end in regulation with a tie to total number of games played tell us the frequency how often team that lose at the end of regulation could have at least draw if attempted a field goal rather than go for 4th down in a field goal territory and score 3 points. Ratio is that circa every 72nd game played was affected by this phenomena that losing team could at least tie the game at the end of the regulation if would not take the risk.

3.1.3 Results

Interesting it is also to look individually at teams which contribute to this total number of games affected by the phenomena most. On the bar chart first two positions are Los Angeles Chargers and Baltimore Ravens what is actually not big surprise because these teams are considered really aggressive with taking gamble of playing 4th down.

Since there are few of observations for each team there can not be any clear conclusions from this frequency but it should show some tendencies of the teams in general and mainly give information of how many games each teams could have at least not lose in regulation if was not too aggressive in play calling.

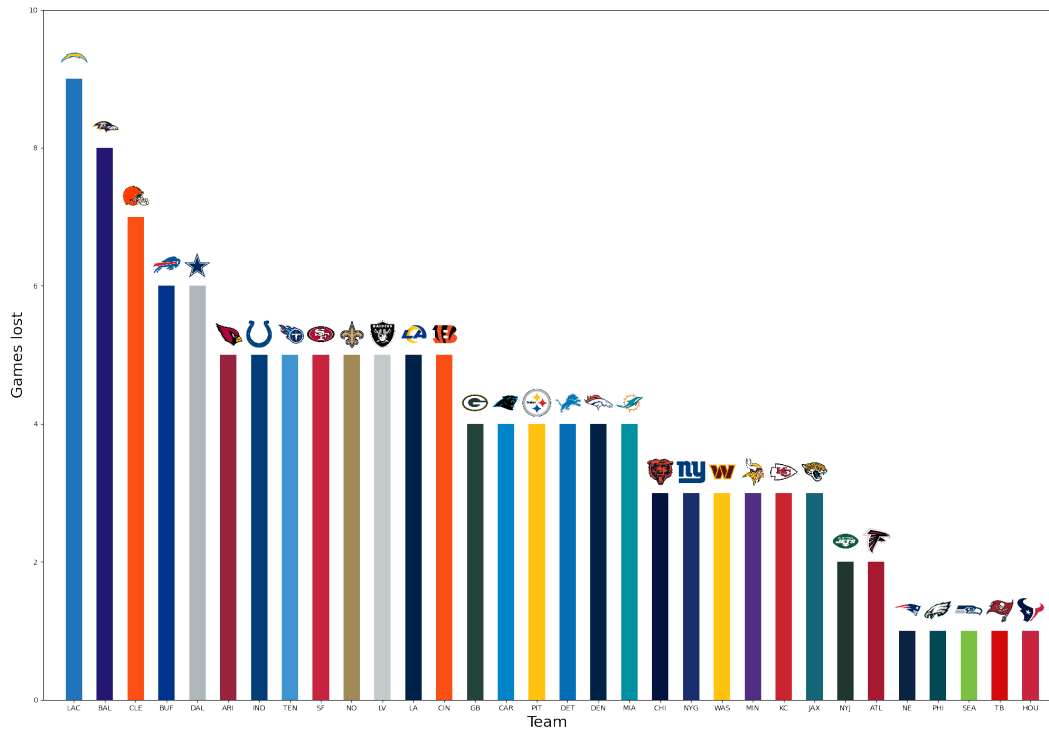


Figure 3.3: Games lost in lower score difference than not converted 4th down with field goal (own processing using Python)

3.2 Freezing the kicker

3.2.1 Specification

In American football icing the kicker or freezing the kicker means practice to call a timeout immediately prior to the start of the attempt in order to disrupt the process of kicking a field goal. The main intention is to throw the kicker off of his routine and force him to feel pressure for a longer amount of time. In order to find out if this technique is effective or not it is crucial to look at the cases when the timeout was called before each field goal attempt. The crude hypothesis is to look at the field goal attempts that followed a timeout called by opposing team (not the team with possession) trying to kick a field goal and those which were not. Important factor is also to look at which team take the timeout because in case it is a team with possession thus team trying to make a field goal attempt successful, the kicker who is executing the field goal usually has the information about the following timeout. Because of this additional information he can adapt and prepare to make a good timing to his whole routine which is really important psychologically for his perfect execution.

Thus it is important to filter every field goal that was attempted with prior timeout called and narrow it to the cases when the opposing team without possession called the timeout. Then there are another parameters that should be specified. For a person not oriented so well in American football it is also good to state that timeout doesn't anyhow help the kicker to regenerate after plays because kicker is not a part of the offense during regular plays. Kicker comes to the field specifically in situations that are played by special teams.

First parameter that should have the biggest importance are field goal attempts with not much time left in regular time. This first parameter can not be strictly determined by data only estimated that psychologically it can has biggest influence on the kicker when there are only couple of minutes left. In this time the kicker has in his mind that his team probably may not get a ball into possession again and also even if they get they might not reach good field position enough for repeating an attempt to score. This situation happen in the 4th last period in the regular time and similarly applies for the overtime. Time constraint will be because of that restricted for last 6 minutes of regulation and whole overtime since in overtime there is even bigger chance that team will not get the possession again.

Second important parameter that is mainly psychological is how tied is the game thus what is the difference in score of both teams in the moment of the field goal attempt. Because of the previous assumption that team might not get another chance we need to narrow biggest score difference into 3 and less points included so in other words situations where successful field goal at least ties the game if not taking possession team into the lead. That is important because otherwise team with a possession even into the field goal territory would not have incentive to kick a field goal and no even tie the game with information that it is very probable after this possession they will no get another chance to score a point. In the cases when field goal does not at least tie the game teams try to score a touchdown by all the possible ways including playing 4th downs and trying to convert them. They actually do not have any other option either kicking a field goal or 4th down not converted is a lost of possession

with not tying a game what has no meaning because they can not win the game. Difference in score at the end of the game is not important in American football when the team lost the game, because at the end of the match counts only who has more points and not by how much they win.

Third condition and restriction is due to possible bias by the extremes with a need to filter outliers. By frequency statistics less probability of making a kick good thus where the number of failures from specific distances is bigger than success ones is direct factor that can misrepresent the effect of taking a timeout. In other words taking a timeout before very distant kick that is very probable not to be successful and can be only desperate attempt for team that has nothing to lose do not affect the probability of the kick in same size as if it was in less distant kicks. Scoring from desperate field goal attempt is unlikely even without taking a timeout before the kick.

Since this parameter is quantifiable best practise seems to be just determine a threshold and consequently filter distances that are above. Determination of the maximal value can be done by looking at the proportions of field goals that were scored and those which were not for every kick distance. Smallest field goal attempt distance where proportions of not scored field goal is bigger than those that were scored can be the threshold behind which we should trim the values.

3.2.2 Procedure

Because of better imagination it is good to look at the total count of a field goals attempted in all games and calculate number of average field goal attempts in each game. There is a median of 4 field goal attempts per a game no matter if they were successful (made) or unsuccessful (missed or blocked combined). For an analyses if freezing a kicker is an effective tactical practise we need to in first place filter situations where there was a timeout called before an execution of a field goal.(7) From all field goal attempts in the count of 23,362 only 2,994 were preceded by timeout what makes circa 12.8 percent of total amount of field goals executed in the given period. Important is also filter if the timeout was called by opposing team. Timeout of team with possession can indicate need for time to decide if field goal will be executed or any other option such as going for a touchdown in this drive rather than attempting a field goal. Additionally the kicker usually has the information about planned timeout of his team so it should not has any psychological influences. This condition narrow our group of field goals attempts from 2,994 to 2,874. From this small difference we can deduct that timeouts before a field goal are hardly ever taken by possessing team. Very important aspect is in which time period of the match is field goal being attempted since timeouts are quite scarce and coaches do not want to waste them on non critical situations. That is a reason why most freezing the kicker tactic is being used on the end of halves most into the two minute warning time interval. There it is needed to just set boarder which is psychologically implied for expectation of not getting another chance. Good practise should be to set this boarder to 6 minutes left in the regulation during 4th period and for the whole overtime. Another set parameter is that the score differential in moment of field goal is

smaller than 3 including. Small score difference implies more pressure on the kicker while attempting a field goal which comes from an importance of the moment.

Parameter that can be quantifiably set is boarder behind which is scoring from field goal attempt so unlikely that it can bias the effect of taking a timeout. Most simple definition can be from the bar chart created with outcomes for every specific kick distance of field goals. As we can see on the graph first time when it is more likely to not score points from field goal is distance of 57 yards thus maximal kick distance that is going to be taken into account is 56 yards highlighted with yellow lines on the chart that shows distribution of field goals attempts in proportion to frequency of attempts illustrated by the bars width.(8)

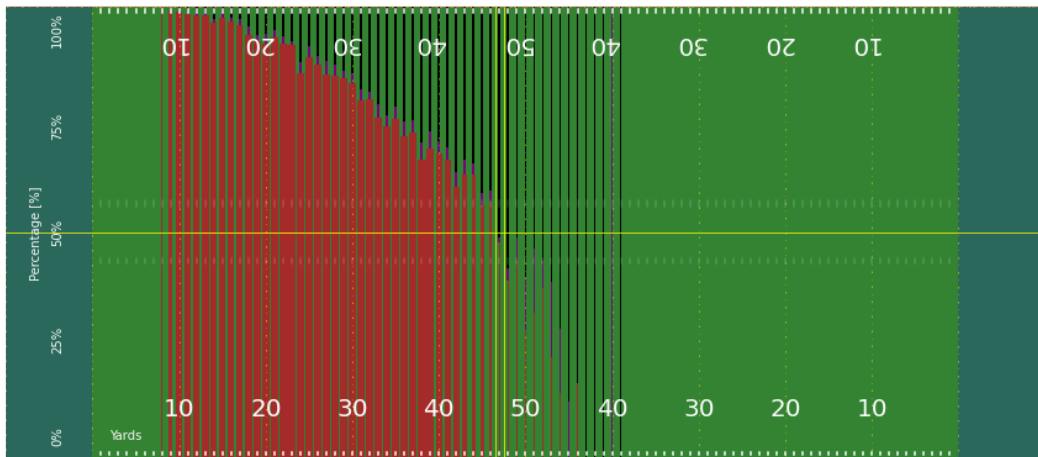


Figure 3.4: Field goals distribution (own processing using Python)

3.2.3 Results

For the evaluation of hypothesis seems best suited chi.squared test of proportions. On the first sight we can see that the proportions differ but for evaluation it should be also statistically significant difference.

Assumptions of the test are met as both variables are categorical (made or not made (missed plus blocked) and mutually exclusive since every field goal attempt can be marked as made or not made and none of the field goal can be both made and not made at the same. All observations of field goal attempts are independent because each field goal is executed individually and volume of observations in both category is far higher than 5. As we can see from output lower in case of field goals without freezing it is 1087 attempt from which 907 were made making rate of scoring from the field goal attempts circa 0.834. Number of field goals that were freezed is in total 793 while 627 were made what makes a proportion of circa 0.791 from total number of attempts. In both groups number of observations is far higher than 5.

Null hypothesis for the test is that these two groups have same proportion and alternative is that they differ thus taking a timeout before field goal to freeze a kicker is significantly different. As we can see on every reasonable significance level we reject the null hypothesis of same proportion in advantage of alternative that proportions are not equal.

From the output of theoretical counts it is apparent that first group should have circa 20 more failed attempts while second group with taken timeouts should have circa 20 less failures to proportions would be the same in both groups.

From the result of the test we can tell that null hypothesis is rejected. That means that there is statistically significant difference between chances of scoring from a field goal in tense situations. It is possible to assume that probability of making field goal is statistically significantly lower when opposing team decide to take a timeout with intention to freeze the opponents kicker.

```
made fieldgoals without freeze = 907.0
made fieldgoals with freeze = 627.0
not made fieldgoals without freeze = 180.0
not made fieldgoals with freeze = 166.0
total fieldgoals without freeze = 1087.0
total fieldgoals without freeze = 793.0
proportion of made fieldgoals without freeze = 0.8344066237350506
proportion of made fieldgoals with freeze = 0.7906683480453972
test statistic = 5.84093586546275
p-value = 0.01565749640060031
hyp made fieldgoals without freeze = 886.945744680851
hyp made fieldgoals with freeze = 200.05425531914895
hyp not made fieldgoals without freeze = 647.054255319149
hyp not made fieldgoals with freeze = 145.94574468085105
```

Figure 3.5: Output of t-test from Python

3.3 Impact of altitude on kick distances

3.3.1 Specification

Important factor of the outcomes that can vary between games and is not strictly defined by the rules is the altitude of stadiums. National Football league rules do not strictly define even particular one specification of stadium in terms of roof or turf. Naturally least enforceable aspect is altitude of stadium that depends on the city altitude. This fact should be factor that enters decision making process as extra parameter. Stadium that is unmatched in this condition by centuries of meters is Denver Broncos home field at Denver, Mile High, Colorado that is in fact one mile high with 1,610 meters above sea level.

Open roof with thin air with density lower in same other conditions fulfilled to others altitudes sets specific physical conditions. When we want to analyse if the ball actually significantly flies further during the games on the Denver Broncos home stadium we need to filter another stadiums as control group. Best independent of other factors situation seems punt since during a punt the kicker is trying to kick a ball as further as it is possible with target to shift opposing team as far as it is possible. This hypothesis is not right in specific situations when kicker is more focused on accuracy of the kick than the length. Situations when accuracy its more important are when punt is being executed in shorter distance to the end zone than length how far is a kicker capable of to kick the ball.(9)

3.3.2 Procedure

When the kicker is closer to the end zone than his maximal punt length he could kick the ball behind the end zone but he want to kick it at least between start of the end zone and 20 yard line which is a starting position of drive following a touchback. Because of this factor it is important to filter observations when biggest effort of the kicker is length and not accuracy of the punt. These situations are where in strict majority it is not even possible to kick behind the opponents end zone without a very auspicious bounce of the ball while kicking it near the goal line is very hard. To set a parameter what is the shortest distance to the end zone when distance is most important while executing a punt it is useful to look at the probabilities of the touchback for each field position that punts were kicked. Then it is only matter of choice what probability is enough for us to ignore option of a touchback.

When creating a groups to compare there is another important parameter to take into account. Denver Broncos home stadium is all over the years an outdoor stadium so the control group of punts should also be from the stadiums that are outdoors. When other types of the stadiums are filtered only difference will stay the altitude and density of air on the field and thus we can compare punts distinguished by the altitude. Important aspect is that since circa half of the punts on the Denver stadium are Denver Broncos punts as home team it could bias the distribution by the Denver Broncos values. Because of that we take into account

only punts of teams opposing Denver Broncos on Denver stadiums thus away teams of all other teams on the Denver stadium. As a control group we look at the punts of the away teams at every other stadium that has much lower altitude which are filtered from Denver Broncos away games. Looking at the only away teams punts prevent distribution from being biased by difference of only away teams on Denver Broncos home stadium and mix of home teams and away teams on every other outdoor stadiums. When all of these conditions are implied we can compare two groups that differ only in the altitude of the environment where were executed.

By the descriptive statistics with calculation of percentage of touchback from the punts from each yard line distance it is possible to deduct that from the distance of more than 69 yards from the end zone more than 99 percent of the punts does not end as touchback so we can count with the effort of the punters to kick as far as they are possible. That is where we can state the border and filter only punts that were in distance of 70 or more yards from the end zone of opposing team. Boarder can be easily spotted on the chart where orange line is representing percentage of punts from each yard distance that did not end as a touchback. Light blue lines determine at what yard line it is firstly rate of more than 0.99 of punts that did not result in a touchback.

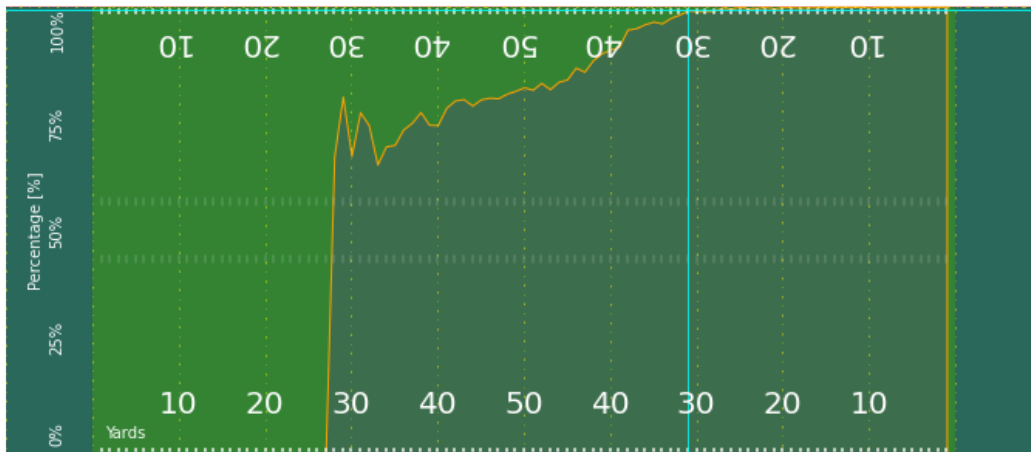


Figure 3.6: Punts without touchback (own processing using Python)

Important is to filter missing values which means touchbacks and can not be measured precisely and 0 values that means blocked punts. For missing values we should just substitute with distance from the goal line and blocked punts should be ignored as it is not related to altitude and the flight of the ball.

Now last filtering is only grouping observations into one group of punt with distance from the end zone bigger than 70 yards on Denver Broncos home outdoor stadium in Denver Mile High Colorado and other outdoor stadiums. Only difference will be the altitude and density of air on the field. Important aspect is that since circa half of the punts on the Denver stadium are Denver Broncos punts as home team it could bias the distribution by the Denver Broncos values. Because of that we take into account only punts of teams opposing Denver Broncos

on Denver stadiums thus away games of all other teams on the Denver stadium. As a control group we look at the punts of the away teams at every other stadium that has much lower altitude which are filtered from Denver Broncos away games.

3.3.3 Results

It is possible now to compare means from these two groups. Comparison should be tested with the Student t-test of means. First assumption of this test is independence of two groups that is satisfied since away teams punts are independent of each other and also across the outdoor stadiums. Normal distribution is not a crucial assumption that needs to be satisfied with this volume of observations. Equality of variances of both groups is not rejected by Levene's test which is less sensitive for non normality of distribution of observations than Bartlett's (while Bartlett's did not reject the null hypothesis as well). Null hypothesis of equal variances between groups is not rejected on any reasonable level of significance with p-value of circa 0.865 as test statistic very close to 0 with value of circa 0.029 as can be seen on the output from Python scipy.stats library.

```
Levene test for equality of variances
Null Hypothesis is that variances differ between groups
test-statistic = 0.029048784876857917
p-value = 0.8646701630409563
```

Figure 3.7: Output of Levene's test from Python

Since all crucial assumptions are met we can perform the Student's t-test. Punts on Denver home stadium mean is circa 50.37 while on every other outdoor stadium it is 46.41 which implies that hypothesis of longer punts on Denver stadium in higher altitude and thinner air is plausible since difference between these two means is almost 4 yards. Null hypotheses for t-test is that means of these two groups are equal. Alternate is one sided alternative that away teams punts from Denver Broncos home stadium are on average longer than from the other outdoor stadiums. Student's t-test for these two means has value of t-statistic circa 7.5 and p.value 5.91e-14 what implies that on any imaginable level of significance we reject the null hypotheses of equal means. With evidence from the test it is possible to assume that punt distances of away teams on Denver Broncos outdoor stadium statistically significantly differ from the punts of away teams on other outdoor stadiums. Because there was an effort to filter every other aspect that can affect kick distance it is possible to think that altitude of stadium has significant effect on flight properties of the ball due to less dense air.

```
punts of away teams in Denver on outdoor stadium = 50.37442922374429
punts of away teams outside Denver on outdoor stadiums = 46.40633842671194
t-statistic = 8.948404270663152
p-value = 2.1507131770404566e-19
```

Figure 3.8: Output of chi-squared test from Python

Conlusion

To sum up this bachelor thesis shows how it is possible through the available data to ex post statistically analyse the outcomes that result from specific decisions in various situations. Obtained findings can be then used in future decision making process for increasing chances of achieving more than it would be possible with worse decisions. Firstly was by exact specification analysed situations where teams played too aggressively meaning risking loss of almost certain points in order to gain more points. From results it was possible to identify teams which venturesome behaviour contributed to the losing of whole game that was close to win. This information could be a useful warning for those teams. Teams with highest count of lost matches by that way should analyse deeper every such a game with awareness that it happens to them more often than to the others. This attitude can be applied in every possible aspect of the game arranged by importance because every team should try to improve their results in such situations absolutely and mainly in relation to the other teams. Secondly was disassembled tactical action of taking a timeout before important event in match when successful execution is expected to be delivered by the opposing team. This phenomena is called freezing or icing the kicker because it is linked with important field goal attempts execution. Freezing is practically giving more time to the kicker to think about the execution and also leaving the kicker under pressure of important moment for a longer time. Again even for this situation it was crucial to specify exactly on which occurrences of field goals it is relevant to look. By this hypothesis coaches staff can take into account to use the timeout when the team would face such a situation. Thirdly last hypothesis was dedicated to other than game explicit factors that are not enforced even by rules specifications. Factor that falls into this area is altitude in which games are played. Thin air can influence decisions of teams that are taking into account that the ball should has tendency to fly in longer distances in comparison with same conditions in lower altitudes with more dense air. Effect of altitude was checked by comparison of punt lengths on outdoor stadiums divided into two groups. First group was every other outdoor stadium that punt was executed and the other group was sticking out Denver Broncos outdoor stadiums that are in one mile height above the sea level. Because punt is very independent event in a game and intent of punter is to kick a ball as far as it is possible this effect could be shown on this example well. By showing on this three examples how every good defined enough aspect of the game could be assessed it is possible to demonstrate an importance of dedication to data evaluation.

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